

Quiz- Curve Sketching - Solutions

Part A: True or False – Place your answer in the space provided on the right.

1. If $f(x)$ is a polynomial of degree 3, then its' graph has one point of inflection. TRUE
2. If $f''(1) = 0$, then $(1, f(1))$ must be a point of inflection. FALSE
3. If $f'(c) = 0$, then f must have either a relative minimum or a relative maximum at $x = c$. FALSE

Part B: Full Solutions

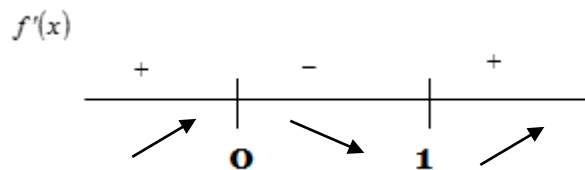
4. Determine any local maximum or local minimum points and the intervals of increase and decrease for the

function $f(x) = \frac{1}{7}x^{\frac{7}{3}} - x^{\frac{4}{3}}$.

$$f'(x) = \frac{1}{3}x^{\frac{4}{3}} - \frac{4}{3}x^{\frac{1}{3}}$$

$$= \frac{1}{3}x^{\frac{1}{3}}(x - 4)$$

$$f'(x) = 0: x = 0, x = 4$$



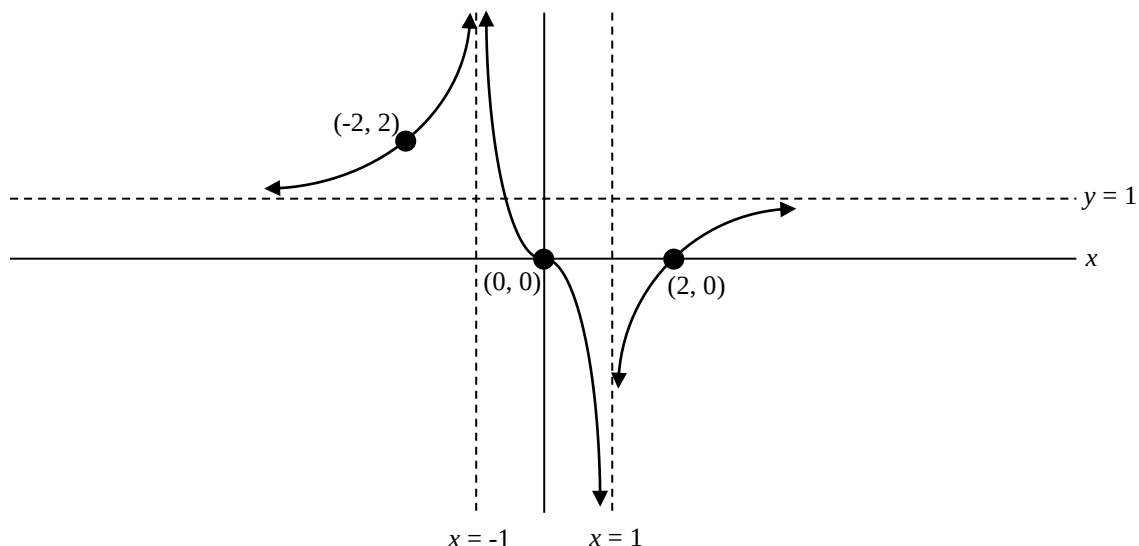
$$f \nearrow: x \in (-\infty, 0) \cup (4, \infty)$$

$$f \searrow: x \in (0, 4)$$

local max(0,0)

local min (4, -2.72)

5. Sketch the graph of a function f such that $f(-2) = 2$, $f(0) = f(2) = 0$, there is a horizontal asymptote at $y = 1$, there are vertical asymptotes at $x = -1$ and $x = 1$, $f''(x) > 0$ for $x < -1$, $f'(x) < 0$ for $-1 < x < 1$ and $f''(x) < 0$ for $1 < x$.



6. For the function $f(x) = 2x^3 - x^4$, determine the critical points and classify them using the second derivative test.

$$f(x) = 2x^3 - x^4$$

$$f'(x) = 6x^2 - 4x^3$$

$$0 = 6x^2 - 4x^3$$

$$0 = 2x^2(3 - 2x)$$

$$x = 0, x = \frac{3}{2}$$

$$f''(x) = 12x - 12x^2$$

$$f''(0) = 12(0) - 12(0)^2$$

$$f''(0) = 0$$

Does not work

$$f''(x) = 12x - 12x^2$$

$$f''\left(\frac{3}{2}\right) = 12\left(\frac{3}{2}\right) - 12\left(\frac{3}{2}\right)^2$$

$$f''\left(\frac{3}{2}\right) = -9$$

Concave Down, Maximum

$$f(0) = 2(0)^3 - (0)^4 = 0$$

$$f\left(\frac{3}{2}\right) = 2\left(\frac{3}{2}\right)^3 - \left(\frac{3}{2}\right)^4 = \frac{27}{16}$$

We cannot classify $(0, 0)$ using the second derivative test since it has failed, but $\left(\frac{3}{2}, \frac{27}{16}\right)$ is a maximum.

7. The graph of the function $f(x) = \frac{ax+b}{(x-1)(x-4)}$ has a horizontal tangent line at the point $(2, -1)$. Find a and b and show that the function has a relative maximum at $(2, -1)$.

$$f(x) = \frac{ax+b}{(x-1)(x-4)}$$

$$f(x) = \frac{ax+b}{x^2-5x+4}$$

$$f'(x) = \frac{a(x^2-5x+4) - (2x-5)(ax+b)}{(x-1)^2(x-4)^2}$$

$$f'(x) = \frac{ax^2 - 5ax + 4a - 2ax^2 - 2bx + 5ax + 5b}{(x-1)^2(x-4)^2}$$

$$f'(x) = \frac{-ax^2 + 4a - 2bx + 5b}{(x-1)^2(x-4)^2}, \text{ horizontal at } (2, -1)$$

$$0 = \frac{-a(2)^2 + 4a + b}{(2-1)^2(2-4)^2}$$

$$0 = \frac{-4a + 4a + b}{4}$$

$$0 = b$$

$$f(x) = \frac{ax}{(x-1)(x-4)}, (2, -1)$$

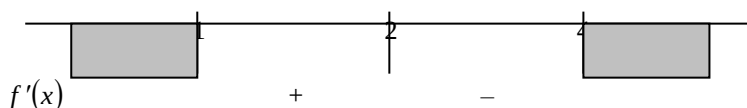
$$-1 = \frac{a(2)}{(2-1)(2-4)}$$

$$-1 = \frac{2a}{-2}$$

$$2 = 2a$$

$$1 = a$$

$$\therefore f(x) = \frac{x}{(x-1)(x-4)} \text{ and } f'(x) = \frac{-x^2 + 4}{(x-1)^2(x-4)^2}$$



Therefore the function has a relative maximum at $(2, -1)$.

8. Given $f(x) = 3x^4 + x^3 - 21x^2$.

a) Find the absolute minimum value of $f(x)$. Show all work.

b) Find the x-coordinate of each point of inflection on your graph of $f(x)$.

$$f'(x) = 12x^3 + 3x^2 - 42x$$

$$= 3x(4x^2 + x - 14)$$

$$= 3x(4x - 7)(x + 2)$$

$$f'(x) = 0: x = 0, x = \frac{7}{4}, x = -2$$

$$f''(x) = 36x^2 + 6x - 42$$

$$f''(0) = -42 < 0 \rightarrow \text{localmax at } x = 0$$

$$f''\left(\frac{7}{4}\right) = 78.75 > 0 \rightarrow \text{localmin at } x = 1.75$$

$$f''(-2) = 90 > 0 \rightarrow \text{localmin at } x = -2$$

$\therefore f(x)$ has absolute minimum of -44 at $x = -2$

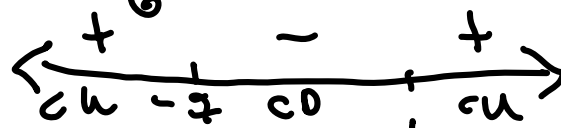
b) $f''(x) = 0$

$$36x^2 + 6x - 42 = 0$$

$$6x^2 + x - 7 = 0$$

$$(6x + 7)(x - 1) = 0$$

$$x = -\frac{7}{6}, x = 1$$



9. Consider the function $f(x) = x^{\frac{4}{3}} + 4x^{\frac{1}{3}}$ that has $f'(x) = \frac{4}{3}x^{\frac{1}{3}} + \frac{4}{3}x^{-\frac{2}{3}}$ and $f''(x) = \frac{4}{9}x^{-\frac{2}{3}} - \frac{8}{9}x^{-\frac{5}{3}}$.

Sketch a properly labelled graph of $f(x)$.

x-int, y = 0

$$0 = x^{\frac{4}{3}} + 4x^{\frac{1}{3}}$$

$$0 = x^{\frac{1}{3}}(x + 4)$$

$$x = 0, x = -4$$

y-int, x = 0

$$f(0) = (0)^{\frac{4}{3}} + 4(0)^{\frac{1}{3}}$$

$$f(0) = 0$$

$$f'(x) = \frac{4}{3}x^{\frac{1}{3}} + \frac{4}{3}x^{-\frac{2}{3}}$$

$$0 = \frac{4}{3}x^{\frac{1}{3}} + \frac{4}{3}x^{-\frac{2}{3}}$$

$$0 = \frac{4}{3}x^{-\frac{2}{3}}(x + 1)$$

$$f''(x) = \frac{4}{9}x^{-\frac{2}{3}} - \frac{8}{9}x^{-\frac{5}{3}}$$

$$0 = \frac{4}{9}x^{-\frac{2}{3}} - \frac{8}{9}x^{-\frac{5}{3}}$$

$$0 = \frac{4}{9}x^{-\frac{5}{3}}(x - 2)$$

	-1	0	2	
$f'(x)$	-	+	+	+
$f''(x)$	+	+	-	+
$f(x)$				

$$f(-1) = (-1)^{\frac{4}{3}} + 4(-1)^{\frac{1}{3}}$$

$$f(-1) = -3$$

$$f(0) = (0)^{\frac{4}{3}} + 4(0)^{\frac{1}{3}}$$

$$f(0) = 0$$

$$f(2) = (2)^{\frac{4}{3}} + 4(2)^{\frac{1}{3}}$$

$$f(2) = 7.56$$

Therefore, the function has a minimum at $(-1, -3)$, a point of inflection at $(2, 7.56)$, and a vertical slope at $(0, 0)$.

